

# XRPL Validator — Daily Engineering Report • 2026-03-22

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Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

## Summary

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- **9 commits** (xrpl-validator 9)

## Engineering changes

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- **feat(viewer): auto peer discovery from TMEndpoints** ( 506923e , xrpl-validator) When a connected peer sends TMEndpoints, we automatically connect to new peers we haven't seen before. Filters out local/IPv6 addrs. Max 15 concurrent peer connections. Peers at hop count  $\leq 1$  only.
- **feat(viewer): live peer count with connected/discovered display** ( 861112a , xrpl-validator) /api/peers endpoint returns connected count, known peers, and max. Header shows PEERS 2/5 (connected/discovered) updating every 5s. Peer panel shows all discovered peers sorted alphabetically.
- **fix: sled storage for sync, peer identity verification, consensus wiring** ( 3a37944 , xrpl-validator) Three incomplete items from assessment, now fixed: 1. Sync writes to sled DB alongside JSONL - sled::Batch for atomic page writes - Binary key/value storage (not hex text) - DB at /mnt/xrpl-data/sync/state.sled/
- **fix(viewer): neural network consensus viz with tab-switch fix** ( 24703dd , xrpl-validator) Restore neural network visualization as the consensus page. When switching tabs and returning, accumulated pulses are now flushed (detects  $>0.5s$  frame gap) to prevent lag burst. SSE events skip queuing visual effects while tab is hidden.
- **feat(ledger): transaction application engine with testnet verification** ( c64f330 , xrpl-validator) Complete transaction engine implementing the XRPL ledger close pipeline: - Keylet system: deterministic 256-bit state keys (3 mainnet-verified) - Sandbox: transactional overlay with snapshot/rollback for tec safety - Transactor trait: preflight  $\rightarrow$  preclaim  $\rightarrow$  apply\_common  $\rightarrow$  do\_apply - Payment: XRP direct payments with account creation, reserve checks - OfferCreate/Cancel: DEX order placement with crossing framework - ...
- **feat(viewer): engine dashboard, tx-type visuals, regression tests** ( 6818946 , xrpl-validator) Metrics page: - XRP fee burn counter with sparkline (total + per-round) - Transaction type breakdown bars (color-coded by type) - Engine coverage card (supported vs unsupported percentage)
- **feat(ledger): complete tx type coverage + engine dashboard + audited** ( eb4a41e , xrpl-validator) Transaction types (33 total, up from 6): - Added: Escrow (Create/Finish/Cancel), Check (Create/Cash/Cancel), PayChannel (Create/Claim/Fund), NFToken (Mint/Burn/CreateOffer/ AcceptOffer/CancelOffer), AMM (Create/Deposit/Withdraw/Vote/Bid/ Delete), SetRegularKey, SignerListSet, DepositPreauth, Clawback, Credential (Create/Delete/Accept)
- **feat: live engine, peer network page, dashboard redesign, identity** ( 343ae01 , xrpl-validator) Live Engine: - Opens sled DB (18.7M+ entries) and applies transactions to real mainnet state in real-time - Decodes binary XRPL objects, deducts fees, writes back to sled - Tracks applied/failed/total stats exposed via /api/engine - XRPL\_NO\_ENGINE env var disables engine while sync runs

# Docs and operations

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- docs: add LICENSE (MIT), README with security warnings, .gitignore ( 90697c1 , xrpl-validator)

## Appendix — commits

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| Time  | Repo           | Commit  | Subject                                                                  |
|-------|----------------|---------|--------------------------------------------------------------------------|
| 00:07 | xrpl-validator | 506923e | feat(viewer): auto peer discovery from TMEndpoints                       |
| 00:11 | xrpl-validator | 861112a | feat(viewer): live peer count with connected/discovered display          |
| 00:41 | xrpl-validator | 3a37944 | fix: sled storage for sync, peer identity verification, consensus wiring |
| 00:59 | xrpl-validator | 90697c1 | docs: add LICENSE (MIT), README with security warnings, .gitignore       |
| 19:38 | xrpl-validator | 24703dd | fix(viewer): neural network consensus viz with tab-switch fix            |
| 19:38 | xrpl-validator | c64f330 | feat(ledger): transaction application engine with testnet verification   |
| 20:02 | xrpl-validator | 6818946 | feat(viewer): engine dashboard, tx-type visuals, regression tests        |
| 20:49 | xrpl-validator | ebaa41e | feat(ledger): complete tx type coverage + engine dashboard + audited     |
| 22:34 | xrpl-validator | 343ae01 | feat: live engine, peer network page, dashboard redesign, identity       |